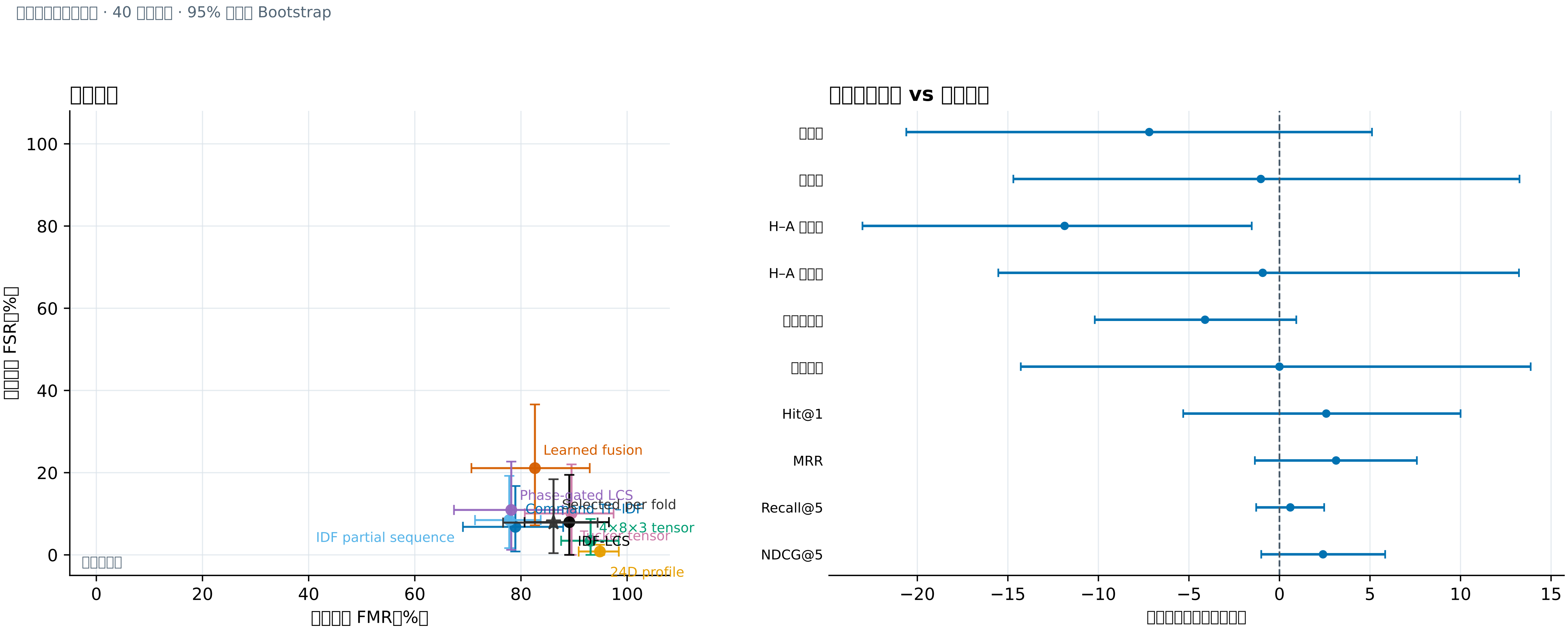
· 40 · 95% Bootstrap
FSR（%）
FMR（%）
0
20
40
60
80
100
Learned fusion
Phase-gated LCS
Command TF-IDF
Selected per fold
4×8×3 tensor
Tucker tensor
IDF-LCS
IDF partial sequence
24D profile
vs
H–A
H–A
Hit@1
MRR
Recall@5
NDCG@5
−20
−15
−10
−5
0
5
10
15